BindCraft: design the interface directly

One-shot design of functional protein binders with BindCraft

Pacesa, Nickel, Schellhaas et al. — Nature 646, 483–492 (2025)

Can a structure predictor become a practical generator of functional binders?

Pacesa et al., Nature (2025), doi:10.1038/s41586-025-09429-6

The question: can small experimental screens be enough?

Traditional binder discovery often requires immunization, library screening, or extensive experimental optimization.

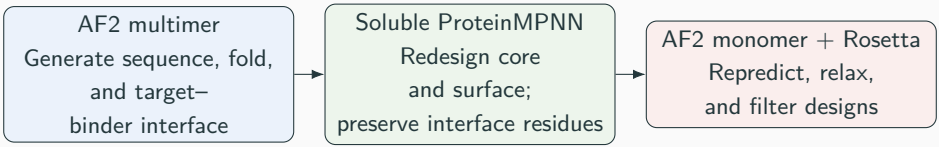
Desired input and output
Target structure + optional binding-site residues $\longrightarrow$ a small set of new protein sequences that bind and change function.

BindCraft asks whether AF2 can help create an interface, rather than only filter interfaces created elsewhere.

The key practical claim is high hit rates among selected designs across diverse targets. It is not guaranteed success for every single generated sequence.

Main; Fig. 1; Conclusions.

Three models play different roles



AF2 weights stay fixed. The optimized object is the binder sequence.

The target complex is repredicted during design, allowing target side-chain and backbone movement within template constraints.

Fig. 1a; Methods: BindCraft design protocol; Supplementary Text, Under the hood.

Running example: design a PD-L1 binder

Actual campaign inputs

- AF2-predicted PD-L1, residues 18–132.
- Target hotspots: 54, 56, 66, 115.
- Binder length sampled within 65–155 residues.

What the algorithm does Start from a random binder sequence, predict its complex with PD-L1, and improve sequence probabilities to favor a folded binder and confident interface.
Hotspots guide the contact loss; they are not hard geometric constraints.

For illustration, a 100-residue candidate has 100×20 adjustable amino-acid logits. This example size is explanatory, not a reported trajectory.

Supplementary Table 1; Supplementary Text: Target preparation & Hotspot selection.

How gradients become an actual protein sequence

For binder length L, let $Z \in \mathbb{R}^{L \times 20}$ contain sequence logits.

$$Z \longrightarrow \text{sequence representation} \longrightarrow \mathrm{AF2}(\text{target}, Z) \longrightarrow \mathcal{L} \qquad Z \leftarrow Z - \eta \nabla_Z \mathcal{L}.$$

Stage	Purpose and default iterations
Continuous (75)	Blend logits with softmax; reject low-confidence starts after 50.
Softmax (45)	Lower temperature to concentrate probability on fewer residues.
Straight-through (5)	Forward pass sees one-hot sequence; gradients use softmax.
Discrete (15)	Test sampled mutations and retain improvements.

Methods: BindCraft design protocol. Models randomly switch among five AF2 multimer weight sets.

The objective combines confidence and physical plausibility

Design pressure	What it encourages
pLDDT, interface pTM	A confidently folded binder and relative chain placement.
Within/between-chain pAE	Consistent internal and interface geometry.
Binder/target contacts	A packed binder with contacts to the chosen target region.
Radius of gyration	Compactness rather than extended chains.
Helicity term	Tune secondary structure; default discourages excessive helices.
Optional termini loss	Nearby N/C termini for insertion into another protein.

These are surrogate objectives, not measured binding free energy. A high-confidence interface can still fail experimentally.

Methods (nine loss components); Extended Data Fig. 1b; Supplementary Fig. 1b–d.

Self-consistency checks reduce model-specific solutions

1. Switch AF2 multimer models during optimization to reduce overfitting to one weight set.
2. Preserve residues within 4 Å of the target; use soluble ProteinMPNN to redesign the remaining sequence (20 sequences per trajectory).
3. Repredict complexes with two template-based AF2 monomer models in single-sequence mode; also predict the binder alone.
4. Relax models and evaluate interfaces using Rosetta.

Rationale: an interface that survives a new sequence context and a different prediction setting is more credible.

These checks are complementary, but do not constitute an independent physical experiment.

Methods; Supplementary Text: Under the hood.

Filtering is stringent; ranking does not estimate affinity

Prediction filters

- Complex pLDDT > 0.8.
- Interface pTM > 0.5.
- Normalized interface pAE < 0.35.
- Bound/unbound binder RMSD < 3.5 Å.

Interface/sequence filters

- Shape complementarity > 0.60.
- > 3 hydrogen bonds; < 4 unsatisfied hydrogen bonds.
- Surface hydrophobicity $< 35\%$.
- Fewer than 3 interface lysines/methionines.

At most two sequences per trajectory pass, encouraging interface diversity. Rank by interface pTM to prioritize testing; this metric does not correlate with affinity in the authors' analysis.

Methods; Supplementary Fig. 6; Supplementary Text: Filtering and Design selection.

“One-shot” still includes substantial computational sampling

Typical selection described in Methods
Generate at least 100 designs passing filters, often from 300–3,000 trajectories; usually test 10 designs selected from the top 20 by interface pTM.

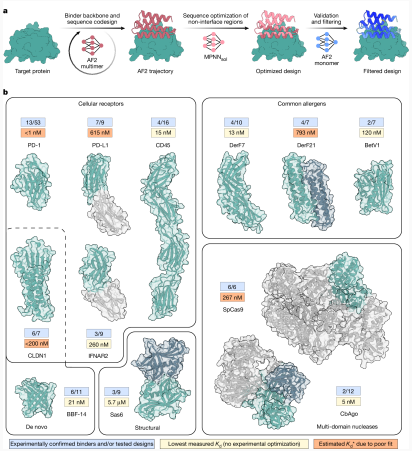
The supplement describes broader practical ranges: difficult campaigns may need 1,000–10,000 trajectories, hours to weeks on one GPU, or yield no accepted design.

Reported hardware examples: about 550 total residues on 32 GB GPU memory; about 950 on 80 GB. These are publication-era guidance, not guaranteed capacity.

Three distinct denominators: initial trajectories, computationally accepted sequences, and experimentally tested designs.

Methods; Supplementary Text: Hardware considerations and Getting started.

The benchmark spans 12 biochemical targets



Pacesa et al., Nature 2025, Fig. 1 (whole, unmodified); CC BY-NC-ND 4.0. Blue: hits/tested; yellow/orange: best measured/apparent affinity.

Receptor binding: success varies by target and assay

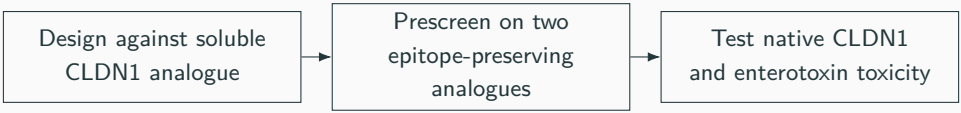
Target	Hits/tested	Best reported affinity	Qualification
PD-1	13/53	$K_d^* < 1\,\text{nM}$	Bivalent Fc fusion; avidity and very slow dissociation.
PD-L1	7/9	$K_d^* = 615\,\text{nM}$	Best binder dimeric in solution; 1:1 target engagement.
IFNAR2	3/9	$K_d = 260\,\text{nM}$	Best binder monomeric; competition with IFNA2.

Competition assays support overlapping epitopes. Selected binders did not cross-bind the tested receptor panel despite similar folds.

Interpretation: selected designs often bind, but neither affinity nor broad proteome specificity follows from this table.

Fig. 2; Extended Data Fig. 2; Supplementary Fig. 3. K_d^* denotes apparent/estimated values.

Claudin: connect soluble-target binding to cell protection



Six of seven designs bound both soluble analogues. Binders 9 and 12 reduced CpE-induced death in CLDN1-expressing cells; both arose from the same trajectory/interface.

MST supports native CLDN1 binding and competition with CpE. The binders did not protect CLDN4-expressing cells. Cytotoxicity plots use $n = 2$ replicates.

Fig. 2i–l; Supplementary Fig. 4; Methods: Cytotoxicity assay (Sf9 cells expressing claudins).

Structure validation: an interface can match the design

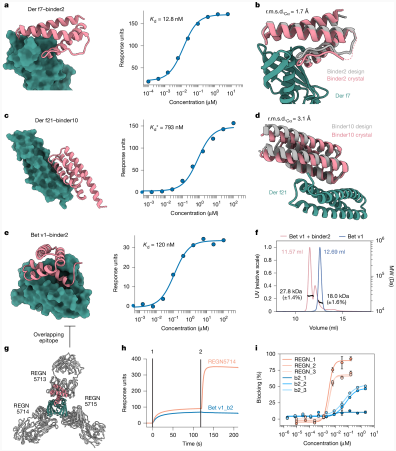
Complex	Binding	Binder RMSD	What is established
BBF-14 / b4	20.9 nM	1.7 Å	Designed interface on a de novo beta-barrel target.
Der f7 / b2	12.8 nM	1.7 Å	Crystal structures at 2.2 and 3.0 Å.
Der f21 / b10	Apparent 793 nM	3.1 Å	2.6 Å crystal structure; local interface discrepancy.

RMSD is binder backbone Cα, after alignment on the target.

The Der f21 discrepancy is attributed to a different interface tyrosine rotamer. Close overall agreement permits meaningful local errors.

Extended Data Fig. 3c,d; Fig. 3b,d; Extended Data Table 1.

Allergens: structure, binding, and patient-serum competition



Pacesa et al., Nature 2025, Fig. 3 (whole, unmodified); CC BY-NC-ND 4.0. Panel i: three donors, two technical replicates.

Bet v1: partial epitope masking, not clinical efficacy

Bet v1 binder2 bound at 120 nM and competed with REGN5713, consistent with the intended epitope.

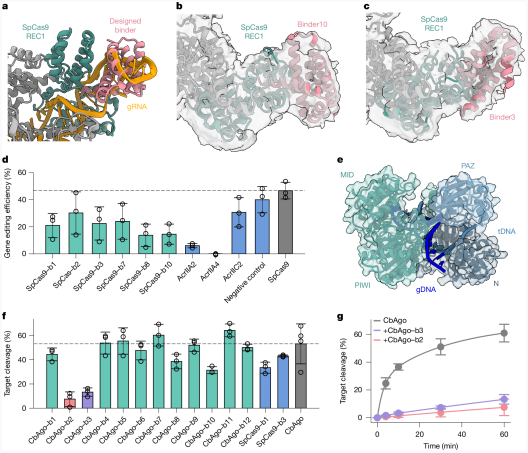
Functional evidence
In a blocking ELISA, the binder prevented up to 50% of allergen binding to serum IgE in two of three donors. The three-antibody REGN cocktail blocked up to 90%.

Allergen was preincubated with binder before exposure to captured IgE. This tests molecular competition under assay conditions.

Limits: small donor sample, technical duplicates, partial blocking, and no demonstration of reduced allergic symptoms or in vivo safety.

Fig. 3g–i; Methods: Birch allergen blocking assay; Extended Data Fig. 4f,g.

Nucleases: designed proteins inhibit biological activity



Pacesa et al., Nature 2025, Fig. 4 (whole, unmodified); CC BY-NC-ND 4.0. Functional plots: $n = 3$, error bars are SD.

What the nuclease experiments establish

SpCas9

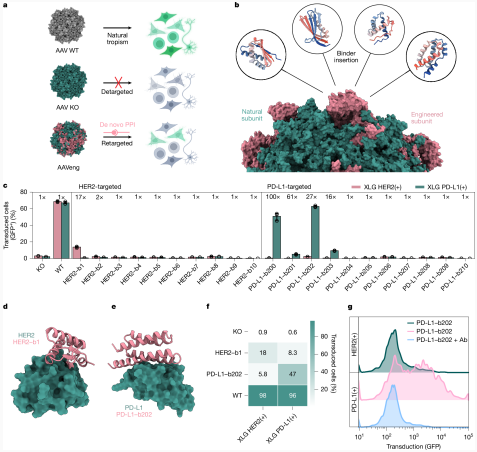
All six tested binders bound apo enzyme. Designed REC1 binders reduced editing in HEK293T cells, compared with no binder. Cryo-EM density supported the location, but local resolution prevented an atomic model. Natural AcrIIA2/4 inhibited more strongly.

CbAgo

Two of 12 designs strongly inhibited cleavage. At 2 μM binder and 0.4 μM enzyme, reported k_{cat} fell about 80-fold (b2) or 40-fold (b3). Binder2 affinity: 5 nM. Guide DNA destabilized its complex with CbAgo, supporting an overlapping binding site.

Fig. 4; Extended Data Fig. 5; Supplementary Fig. 5. Cas9 cellular editing and CbAgo in vitro cleavage are different endpoints.

AAV: move directly from binder design to delivery testing



Pacesa et al., Nature 2025, Fig. 5 (whole, unmodified); CC BY-NC-ND 4.0. GFP transduction in engineered HEK293 cells.

AAV controls connect receptor engagement to transduction

Design binders with nearby termini and insert them into AAV6 between residues 497/498, with GSG linkers. Knockout mutations reduce natural glycan attachment.

1. Compare HER2- and PD-L1-overexpressing cells: one HER2 and four PD-L1 variants showed enhanced target preference in the initial assay.
2. Validate selected HER2-b1 and PD-L1-b202 at equal input: MOI 10^5 vector genomes per cell.
3. Block the PD-L1 binding site with antibody: transduction decreases, supporting receptor-mediated targeting.

Scope: engineered cultured cells, not demonstrated tissue-selective delivery in animals. Initial supernatant screens also mix production and entry effects.

Fig. 5c,f,g; Extended Data Fig. 6; Methods: AAV engineering.

Strong practical evidence, incomplete comparative evidence

- Authors report target-level experimental success of 10–100%, mean 46.3%, among selected designs; targets and assays differ.
- RFdiffusion comparison measures compute time to 100 *computationally accepted* designs, with each pipeline's specified filters. It is not a matched prospective experimental affinity benchmark.
- AF2 monomer filtering can reject real binders; high AF3 confidence also leaves false positives. Interface pTM prioritizes binding plausibility, not affinity.

Our interpretation: the main advance is a useful integrated workflow with functional demonstrations; the paper does not establish universal superiority for every target.

Conclusions; Methods: Computational benchmarks; Extended Data Figs. 1c and 7; Supplementary Fig. 6.

Discussion: what would make “one design–one binder” credible?

Take-home result
Direct AF2-based interface optimization, followed by sequence repair and stringent checks, yields experimentally useful binders from small selected panels.

1. Would a prospective target panel with fixed compute budget and predefined selection rules reproduce the reported success?
2. How much comes from target flexibility, multimer optimization, MPNN, and filtering? Which ablations need experimental validation?
3. Which extra tests are needed for our application: proteome specificity, glycosylation, delivery, immunogenicity, or in vivo function?

Discussion prompts are our interpretation. Sources: full paper and Supplementary Information. Code: github.com/martinpacesa/BindCraft.